Sahar Darafsh



Machine Learning Engineer

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SUMMARY

MSc in Artificial Intelligence and Robotics (Amirkabir University of Technology, 2020), first author of a computer vision paper on arXiv, reviewer of deep learning projects in medical image analysis from 2020 to 2022. Along the way I saw that a good model is only half of the product. The other half is software that keeps running, so I spent 4.5 years as a Java engineer in Munich learning that side properly: TDD, CI/CD, code quality gates, Docker. In 2026 I put the two together: my thesis system rebuilt as an open-source Python project, model on Hugging Face (**98.84% test accuracy**), package on PyPI, CI on three operating systems. After that, a sentiment service for journal entries in English and Persian, with Transformers models and a local LLM, served over gRPC and tracked in MLflow. AI is what I care about, and now I can also ship it.

SKILLS

ML / Deep Learning: Keras · TensorFlow · PyTorch · CNN training · fine-tuning · evaluation (accuracy, F1, confusion matrix) · Hugging Face Hub

Computer Vision: OpenCV · video/motion-based feature encoding · image classification · activity recognition · background subtraction · embedded CV design (Xilinx ZCU102, thesis)

NLP & LLMs: Hugging Face Transformers · XLM-RoBERTa · zero-shot classification (NLI) · local LLMs (Ollama, Qwen) · prompt engineering · sentiment analysis · multilingual text (English, Persian)

Python & Tooling: Python · pandas · Git · pytest · ruff · CLI design · layered architecture · automated unit testing

Backend, MLOps & DevOps: Java · TypeScript · NestJS · REST APIs · gRPC · Protobuf (Buf) · MLflow · Spring Framework · Next.js · React · Docker · Kubernetes · CUDA · AWS · CI/CD (Jenkins, GitHub Actions, GitLab CI) · SonarQube · MySQL · PostgreSQL · TDD/JUnit

RESEARCH & PROJECTS

Daily Journal Sentiment Analysis (private GitLab repos)

Medium write-up

- Reads journal entries in English or Persian and decides if the day was good, neutral or bad, and why.
- Compared a zero-shot model, a sentiment model and a local LLM on my own labels (accuracy, speed, GPU use).
- Runs as an API service that other apps call. Every new version is built and published automatically.

LLM Listing Assistant (private GitHub repo)

- Full stack app: a seller describes changes to their electric car in a chat, a local LLM (Qwen 2.5 on Ollama) turns the text into suggested edits, and an employee confirms or rejects each one.
- Finds the VIN with a regex and asks the LLM for JSON with only the allowed fields (mileage, battery health, equipment). One message can hold several edits, and equipment can be added or removed.
- NestJS and TypeScript REST API with TypeORM, PostgreSQL migrations and Swagger docs. Next.js and React frontend. Docker images pushed to GHCR by GitHub Actions and deployed on Kubernetes.

adaptive-motion-preprocessing: open-source Python package

github.com/sdrfsh/adaptive-motion-preprocessing · pypi.org/project/adaptive-motion-preprocessing · Medium write-up

- Python package that turns video frames into same-size motion images for neural networks (older frames dim, newer ones bright), with motion triggering, noise reduction and background subtraction as pluggable stages.
- Released on PyPI. CI runs the tests on Linux, Windows and macOS, and every release has to pass it first.

Real-Time Activity & Intention Recognition: MSc thesis to open-source system

github.com/sdrfsh/realtime-activity-and-intention-recognition · huggingface.co/sdrfsh/alexnet-door-entry-classifier · arxiv.org/abs/2107.12744 · Medium write-up

- Built an end-to-end video-to-decision pipeline: compresses a motion clip into a single time-encoded image, classified by a lightweight AlexNet-style CNN. Designed for embedded use: the CNN and image pipeline were simulated with Xilinx simulators for a ZCU102 FPGA board.
- Created the training dataset from scratch (no public dataset existed). Accuracy in the paper: **98.78%** on it, **97.95%** on KTH, 100% on Weizmann, 78.48% on HMDB-51. First author of the arXiv preprint.
- Rebuilt the system in 2026 as a tested Python codebase with a CLI for data prep, training, batch prediction and live webcam prediction that runs locally, without the cloud. The released model on Hugging Face reaches **98.84% accuracy** and **F1 0.99** on the held-out test set.

WORK EXPERIENCE

Full Stack Developer · Convien GmbH, Munich | Mar 2025 – Present

- Build and maintain full-stack features for a large-scale meeting-place optimizer in Java and Apache Wicket, keeping high availability and sub-second response times.
- Lead code-quality initiatives against code smells, vulnerabilities, and technical debt.
- Own quality management and release testing.

Software Developer · TESTIFI GmbH, Munich | Mar 2022 – Mar 2025

- Shipped an Android app in Flutter, then moved to backend Java to build QualityOps, a tool for evaluating and visualizing software quality metrics.
- Built core QualityOps features with TDD from stakeholder requirements.
- Contributed to CAST, a cross-team test-automation initiative.
- Employee of the Month (Mar 2024) for fast, high-quality bug fixes.

Deep Learning Mentor and Teaching Assistant (Part-Time) · Maktabkhooneh, Tehran | Jun 2021 – Mar 2023

- Taught and mentored 100+ students in Reinforcement Learning, CNNs, and Deep Learning.
- Co-developed the course curriculum with industry examples.

AI/ML Project Reviewer · Abadan University of Medical Science, Abadan | Jun 2020 – Mar 2022

- Reviewed deep-learning projects in medical image analysis (model architecture, training procedure, evaluation metrics).

EDUCATION

MSc, Artificial Intelligence and Robotics · Amirkabir University of Technology, Tehran | Sep 2017 – Jun 2020

Thesis: "Designing an Embedded System for Activity Recognition Using a Surveillance Camera" (see Research & Projects for the resulting repo, paper, and model)

BSc, Computer Engineering · Isfahan University of Technology, Isfahan | Sep 2011 – Sep 2016

CERTIFICATES

- AI for Medicine Specialization, DeepLearning.AI (Aug 2021)
- Deep Learning Specialization, DeepLearning.AI (Jun 2021)
- Reinforcement Learning Specialization, University of Alberta (May 2021)
- Generative Adversarial Networks (GAN) Specialization, DeepLearning.AI (May 2021)

LANGUAGES

English (Fluent) · German (B1 telc certificate, B2 in progress) · Persian (Native)